

$$\begin{array}{ccc}
 (f, g) & \xrightarrow{\quad \star \quad} & f \star g \\
 \mathcal{F} \times \mathcal{F} \downarrow & & \downarrow \mathcal{F} \\
 (\hat{f}, \hat{g}) & \xrightarrow{\quad \cdot \quad} & \hat{f} \hat{g}
 \end{array}$$